DPE-Based Neuromorphic Hardware for Continuous-Time Biomedical Signal Classification

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***Abstract*—Accurate diagnosis of pathological conditions from biological signals such as electrocardiograms (ECGs) is often performed offline, making it time-consuming, costly, and inefficient, particularly when abnormal patterns are rare and long-term monitoring generates large amounts of data. To address this, the paper proposes a compact, scalable, and programmable neuromorphic system designed for real-time preliminary diagnosis. The proposed design processes ECG signals using a delta-modulator-based spike encoder, followed by classification with a dot-product engine (DPE)-based spiking neural network (SNN) processor. The architecture supports online learning using spike-timing-dependent plasticity (STDP) and includes a scan-chain programming circuit that allows processor parameters to be reconfigured during runtime. The proposed system has been implemented and validated on four FPGA platforms, demonstrating promising results in terms of resource utilization, power consumption, and operating frequency, with a maximum frequency of 333.3 MHz. The design was subsequently implemented in a 130 nm CMOS process, taking up an area of 95,792 μm^2, and dynamic power consumption as low as 2.62 μW.**

***Index Terms*—Neuromorphic, ECG, spiking neural networks, dot-product engine, delta modulator, FPGA, ASIC.**

I. Introduction

Cardiovascular diseases (CVDs) remain one of the leading causes of mortality worldwide, with a projected trend to remain the main health challenge in the coming years [1]. Early detection of arrhythmia is critical in preventing severe cardiac events, such as strokes and myocardial infarctions [2]. Electrocardiography (ECG) serves as the gold standard for non-invasive cardiac assessment, providing vital real-time information about heart function [3]. However, traditional clinical diagnosis is often performed offline, where long-term monitoring generates massive amounts of data that are time-consuming and costly for experts to review manually.

The recent growth of wearable technologies, such as smartwatches, armbands, and disposable ECG patches—has enabled the continuous monitoring of vital signs in everyday environments [4], [5]. While these devices can capture high-quality electrocardiograms or ECG signals conveniently, the primary challenge lies in the automated, real-time classification of these signals at the edge. Most state-of-the-art diagnostic tools rely on deep learning (DL) models, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), which achieve high accuracy but are computationally expensive and memory-intensive [5], [6]. Because wearable devices are typically battery-powered and resource-constrained, deploying such power-hungry artificial neural networks (ANNs) often necessitates offloading data to the cloud. This approach not only incurs high energy costs for data transmission but also raises privacy concerns and dependency on stable network connectivity [7].

To bridge the gap between high-fidelity monitoring and low-power execution, spiking neural networks (SNNs) have emerged as a promising alternative. Inspired by biological neural processing, SNNs operate in an event-driven, sparsely activated domain where information is processed as discrete spikes [3], [8]. This 3rd-generation machine learning framework is inherently suited for temporal signal analysis, such as ECG, as it mirrors the sparse and event-driven nature of biological heart rhythms. By utilizing neuromorphic computing principles, where memory and processing are collocated, SNNs significantly reduce power consumption and computational overhead compared to traditional von Neumann architectures [3], [9].

Despite the potential of SNNs, many existing implementations struggle with the complexities of real-time signal encoding [3] and the hardware-level reconfigurability required for diverse patient profiles [4]. In this paper, we propose a compact, scalable, and programmable neuromorphic system for preliminary real-time arrhythmia detection using ECG signals. Our architecture utilizes a delta-modulator-based spike encoder and a simple dot-product engine (DPE)-based SNN processor followed by a winner-take-all (WTA) module. Key features include online learning via spike-timing-dependent plasticity (STDP) and runtime reconfigurability through a scan-chain module, allowing parameters to be easily tweaked for other continuous-time biomedical signals as well as ECGs.

The rest of the paper is organized as follows. Section II dives into the detailed description of the system. Section III explains and tabulates the results from different FPGA and ASIC implementations. Finally Section IV concludes the paper.

II. Hardware Architecture

The DPE-based SNN system has been implemented with SystemVerilog for FPGA and ASIC projects as shown in Fig. 1. The architecture takes inspiration from event-driven biological systems to achieve high energy efficiency. Communication

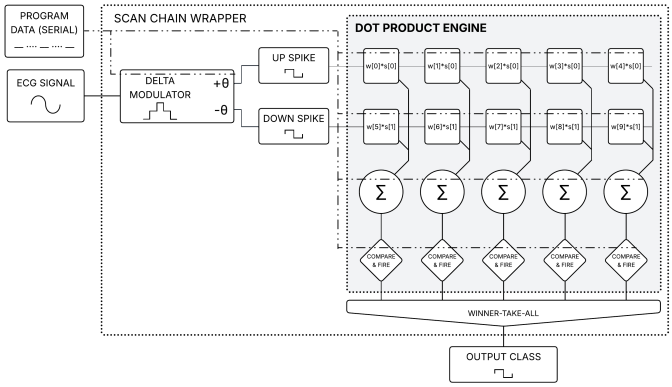

Fig. 1. DPE-based neuromorphic hardware block diagram

is achieved through discrete temporal events, or 'spikes', obtained through a specific spike encoding scheme. ECG analysis promotes this mode of communication, as cardiac signals are rhythmic and can be analyzed by its temporal features [3]. Synapses act as the primary storage element; their weight represents the strength of connection between neurons; the leaky integrate-and-fire (LIF) neuron integrates incoming synaptic stimuli while periodically leaking its membrane potential, until a threshold is reached. The processing of information only occurs when a spike occurs, reducing redundant computations and allowing compact hardware. The following subsections describe the modules that translate these biological ideas into a synthesizable DPE-based crossbar array.

A. Delta Modulator

Delta modulation is a spike encoding scheme that generates a binary spike when the difference between consecutive samples exceeds a configurable threshold. Specifically, the delta modulator generates two spike streams. An 'UP' spike is triggered when the difference is greater than the threshold; a 'DOWN' spike is activated when the difference is less than the threshold. Increasing the threshold makes the delta modulator less sensitive to small changes in the input signal. In contrast, decreasing the threshold effectively increases the resolution of the encoding and spike density. The threshold can be tuned to to match the input data and effectively extract the necessary features of the sampled input signal. This architecture can be easily modified to get more output spikes for a differently sized DPE. This spike encoding scheme has has been shown to be effective in previous works [10].

B. Dot-Product Engine based SNN

The spiking neural network is implemented using a dot-product engine, which is implemented using a neuron array along with a synaptic crossbar. The architecture for this implementation is a 2x5 array. The two rows correspond to the differential spike streams from the delta modulator, while the five columns represent the output classes for ECG classification. The array operates synchronously; each column includes connections to input spikes using synapses that allow online learning computed with a 'neuron time' signal, tracking the time since the last postsynaptic action potential. This existing design can be easily modified to implement different numbers of inputs and outputs in a DPE configuration for different biomedical signal classification applications. In addition to that, multiple DPEs can be connected together to create a deep spiking neural network. The main building blocks of this DPE are the synapse units and the neuron units, each consisting of an accumulator and a compare-and-fire module, which are described in detail below.

1) Synapse Unit: The synapse unit module serves as the main block of the crossbar and serves two functions: inference gating and plasticity logic. The gating logic determines how only the presence of an input binary spike triggers a 4-bit synaptic weight (both positive and negative) contribution to the column accumulator. Additionally, the time since the last input spike (synapse time), or presynaptic action potential, is tracked to contribute to spike-timing-dependent plasticity learning to update synapse weights. Pre-before-post synaptic spikes induce long term potentiation (LTP), or strengthening the synaptic weight; post-before-pre spikes induce long term depression (LTD), or weakening the synaptic weight. If the spikes are closer in time, it results in a higher magnitude of weight change than spikes that are further apart [11]. The magnitude of weight adjustment is determined by a look-up table, indexed by the difference between postsynaptic 'neuron time' and presynaptic 'synapse time', effectively performing as an efficient linear approximation of STDP.

2) Accumulator: The outputs of the synapse units within a column are passed into the accumulator module of the neuron, to perform the summation of the weight/spike products and update the membrane potential. To emulate biological neurons, the membrane potential periodically leaks towards the resting potential when not receiving input stimuli, suggesting that a certain temporal density of spikes is necessary to drive the membrane potential towards the threshold. Furthermore, a short refractory period is applied after the neuron fires, during which the neuron is not responsive to stimulation. During this period, synapses do not contribute to accumulation in the membrane potential [12].

3) Compare and Fire: Each column contains a compare and fire unit, which monitors the membrane potential against a pre-configured threshold. When exceeded, the neuron fires, generating an output spike, the membrane potential is reset to its resting potential, and a refractory period is enforced.

C. Winner-Take-All Circuit

The five output spike signals are passed into a first-spike WTA circuit. Once a single neuron fires within a time-frame, the other columns are inhibited and the membrane potentials are reset. In the context of the entire system, this serves as the final classification decision, which is expressed as a single spike or classification decision.

D. Scan Chain Wrapper

The scan chain wrapper serves as a programming infrastructure for the entire system, providing a serial-input, parallel-load interface for post implementation tuning of several system

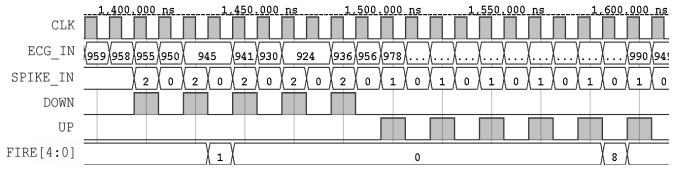

Fig. 2. Behavioral simulation of the proposed hardware in response to ECG signals.

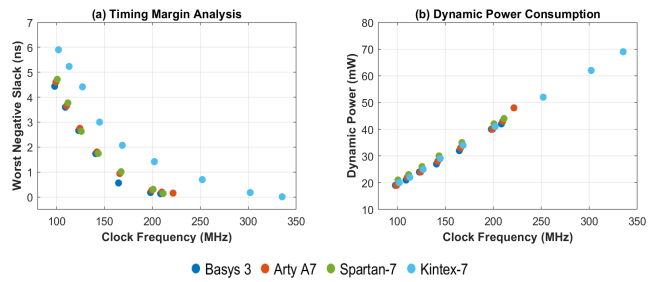

Fig. 3. Analysis showing the change in slack and dynamic Power with frequency on 4 FPGAs.

parameters across each functional unit. This layer utilizes a dual-clock strategy to avoid meta-stability and allow stable tuning, while minimizing the I/O overhead of the system. The wrapper uses a bit packing protocol to load the following system parameters: ten initial synaptic weights, neuron firing threshold, refractory period, and leak rate, making the system runtime reconfigurable. These parameters can be reconfigured to process not only ECGs, but also other continuous-time biomedical signals.

III. Results and Discussion

The proposed hardware was simulated with randomized synaptic weights, and a discretized ECG signal from the MIT-BIH Arrhythmia Dataset [13]; this data was digitized at 360Hz with 11 bit resolution over a 10mV range. Fig. 2 illustrates a behavioral simulation timing diagram from Vivado, ran with a 100 MHz clock. The diagram shows the 11-bit *ECG_IN* being processed by the delta modulator with a sensitivity threshold of 4, resulting in *UP* and *DOWN* spike streams. The *SPIKE_IN* signals drive the DPE-based SNN, leading to specific neuron *FIRE* events, shown in the 5-bit bus. The successful generation of an end-to-end classification output from raw physiological signals validates the functional integrity of the signal processing pipeline.

The proposed model was implemented on software for heartbeat classification using the MIT-BIH Arryhthmia dataset [13] using snntorch [14], which is an open-source Python library built on PyTorch for modeling, training, and evaluating spiking neural networks using surrogate-gradient learning. It enables customized spike encoding schemes, neuron and synapse models, and calculates loss functions that enable training of event-driven neural architectures on conventional hardware while remaining compatible with neuromorphic deployment. ECG beats extracted from the MIT-BIH Arrhythmia Database were normalized on a per-beat basis and encoded using a two-channel delta modulation scheme. The network achieved an overall test accuracy in the range of 75.62% to 79.28% using only one DPE layer with $2-4$ input and 5 output neurons with different parameter values, which is promising as the system is very scalable and reconfigurable making it possible to achieve higher performance. The subsections below discuss the hardware performances for this system.

A. FPGA Implemenatation

The proposed design was synthesized and implemented on the following four FPGA boards: Basys 3, Arty A7, Spartan-7, and Kintex-7. The lightweight footprint, timing margin, and energy efficiency of the DPE-based spiking architecture supports the feasibility of this architecture for ASIC implementation in low-power, wearable edge devices. The frequency, power, and resource utilization analysis of this architecture across the 4 FPGA targets is detailed below.

1) Frequency Analysis: To evaluate the performance and scalability of the SNN architecture, a timing analysis was performed on the 4 FPGA boards. By sweeping the global clock frequency and monitoring the worst negative slack (WNS) until failure, the maximum operating frequency of the system for each target FPGA; the resulting slack versus frequency results are shown in Fig. 3(a). The Basys 3, Arty A7, and Spartan-7 boards all reached a maximum operating frequency over 210.52 MHz, while the Kintex-7 reached a maximum operating frequency of 333.33 MHz. This reflects strong timing resilience, as the design maintains a high baseline frequency on cost-effective Artix and Spartan boards and scales efficiently on the higher performance Kintex-7. Given that ECG classification tasks operate at lower physiological sampling rates, a high maximum frequency provides a significant potential for increasing network density through time multiplexing, as well as achieving ultra-low power consumption through power-frequency scaling.

2) Power Analysis: The dynamic power analysis of the end-to-end architecture demonstrates a highly efficient linear scaling profile across all 4 FPGA platforms, reflected in Fig. 3(b). Starting at 20 mW at a frequency of 100 MHz, the dynamic power linearly increases with respect to frequency to a maximum of 69 mW at 333.33 MHz for the Kintex-7. The low power envelope and high-frequency capability of the system allow for aggressive frequency scaling to minimize power consumption, supporting its potential in battery-powered wearable devices.

3) Resource Utilization Analysis: The design synthesized and implemented across 4 FPGAs demonstrates a high resource efficiency; the entire system requires 519 to 527 LUTs and 424 FFs, with room for further improvement. Shown in Table I, the implementation consumes a maximum of 2.50% of the available LUTs and 1.02% of the FFs on the Basys 3 board. Furthermore, the I/O count is limited to 27 pins, due to the integration of the serial scan chain wrapper. The compactness of this design promotes scaling the number of

TABLE I
RESOURCE UTILIZATION COMPARISON ACROSS FPGA TARGETS

Resource Metric	Basys 3	Arty A7	Kintex-7	Spartan-7
LUT	519	527	518	518
Utilization (%)	2.50	0.83	1.26	0.81
FF	424	424	424	424
Utilization (%)	1.02	0.33	0.52	0.33

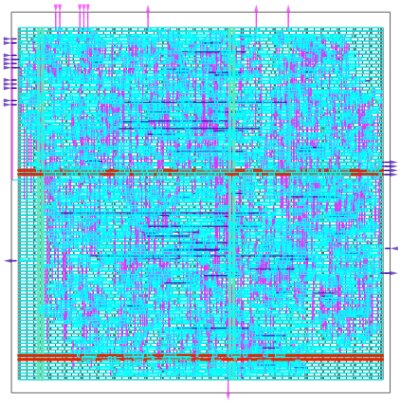

Fig. 4. Post-routing layout of the neuromorphic architecture

synapses and neurons by increasing the size of the crossbar array or cascading multiple arrays for more complex ECG analysis and accurate classifications.

B. ASIC Implementation

The proposed DPE-based SNN system was synthesized and implemented as an application specific integrated circuit (ASIC) using a 130nm CMOS process. The process produced a LVS and DRC clean layout, shown in Fig. 4, while meeting a target frequency of 100 MHz. In addition, timing, power, and area tradeoffs were explored at lower and higher frequencies of 1 MHz and 200 MHz.

To ensure successful routing, the floorplan was constrained to 85,280 μm^2, targeting a utilization of 30%; the final die area is 95,792 μm^2. The architecture boasts a compact logic footprint, with a design instance area of 31,799 μm^2, 31,647 μm^2, and 33,941 μm^2 at 100 MHz, 1 MHz, and 200 MHz, respectively. This footprint promotes the scalability of this architecture. Finally, at 100 MHz, the chip consumes a total power of 7.13 mW, with 2.64 mW of that accounting for switching power. The power scales linearly with frequency, consuming 26.2 µW and 5.62 mW of switching power at 1MHz and 200MHz, respectively.

IV. CONCLUSION

The proposed neuromorphic system offers a highly efficient and scalable solution for real-time arrhythmia detection, bridging the gap between high-fidelity ECG monitoring and low-power edge computing. Using a delta-modulator-based spike encoder and a dot-product engine, the architecture achieves significant energy savings through event-driven processing and localized memory. The integration of online STDP learning and a scan-chain wrapper ensures that the system remains adaptable to diverse patient profiles through runtime reconfigurability, which also enables the processing of other biomedical signals other than ECGs. Validated across multiple FPGA platforms and a 130 nm CMOS process, the design demonstrates great performance, reaching operating frequencies of up to 333.3 MHz while maintaining a compact silicon footprint of 95,792 μm^2. These results underscore the potential of this neuromorphic approach for the next generation of battery-powered, intelligent wearable medical devices.

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